

A GUIDE TO PRODUCT QUALITY STANDARDS LEATHER & LEATHER PRODUCTS



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ABBREVIATIONS

ACCC	Australian Competition and Consumer Commission
ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
AS	Australian Standards
ASTM	American Society for Testing and Materials
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
CPSIA	Consumer Product Safety Improvement Act
CPSA	Consumer Product Safety Act
EC	Council Regulation (Regulation of the European Parliament and Council)
ECHA	European Chemical Agency
EN	European Standards
EU	European Union
FHSA	Federal Hazardous Substances Act
GCC	Gulf Co-operation Council
GSO	GCC Standardization Organization
IS	Indian Standard
ISI	Indian Standards Institution
ILAC	International Laboratory Accreditation Co-operation
ISO	International Organization for Standardization
MLA	Multilateral Recognition Arrangement
MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
SASO	Saudi Standards, Metrology and Quality Organization
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2,40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged

Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This compendium brings relevant regulations and standards pertaining to the leather industry and the various products manufactured in different districts of the state. Uttar Pradesh is renowned for its leather goods and contributes to nearly 32% of the national leather production. The state has a large bovine population which has helped it emerge as a leading producer of leather. Kanpur and Agra are acclaimed for production and export for leather and leather products in Uttar Pradesh. The testament to this is the GI tag award to the Leather Saddles produced in Kanpur.

LEATHER AND LEATHER PRODUCTS, KANPUR NAGAR, AGRA, UNNAO, HAMIRPUR

1. INTRODUCTION

Leather making is one of the oldest crafts (>3000 years) known to man. The process converts Hides and skins to become durable and capable of being used for a wide range of purposes. The raw material used for making leather is a by-product from the meat processing industry. The varied uses of leather demand different properties with Leather being used to make a variety of items, including clothing, footwear, handbags, furniture, tools and sports equipment. Properly cared for, Leather can last for decades.



Leather processing encompasses four major unit operations like pre-tanning, tanning, post tanning, and finishing. The primary processing sequences of leather is shown in Figure 1.

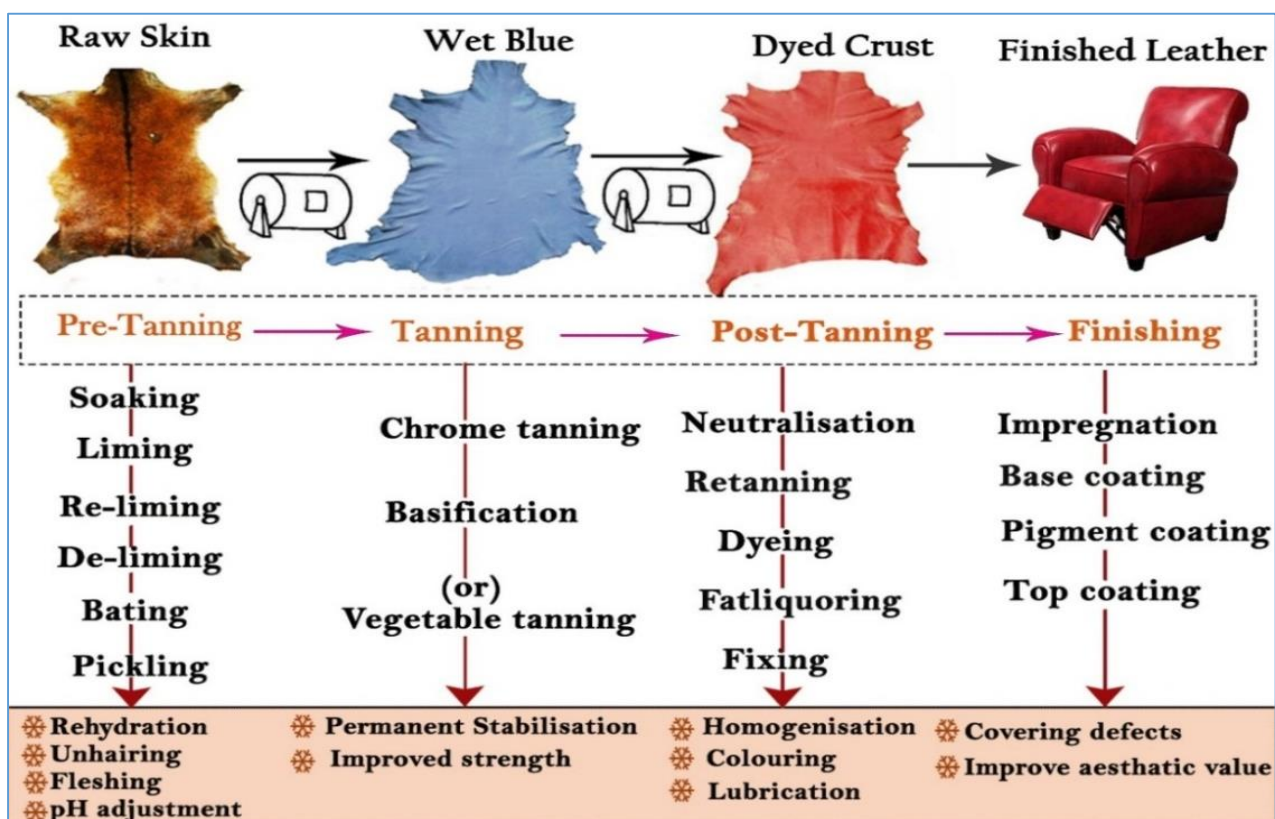


Figure 1: Overview of Leather Processing

2. RAW MATERIALS

The raw material required for preparation of leather and leather products is primarily cattle hides. A wide variety of hides are also obtained from other animals such as horses, pigs, sheep and goat. Other raw materials required include:

- Salt
- Milk of Lime and other chemicals used as sharpening agents like sodium sulphide
- Sulphuric acid
- Chromium Sulphate

3. MANUFACTURING PROCESS

3.1. Leather Processing

3.1.1. Sorting and Trimming

Hides and skins are sorted into several grades based on size, weight, and quality. Trimming is generally carried out during the sorting process. The edges (legs, tails, face, udders, etc.) and the unwanted portions of the raw hides and skins are cut off.

3.1.2. Desalting

Desalting is the removal of salt from the salted stock by gently beating the hide or skin on a specially constructed masonry platform of two feet height with a curved surface on top to prevent mechanical damages during manual desalting operation in the conventional process. In addition to this, desalting can also be carried out in specially designed drums. By drumming the salted goods, it is possible to remove about 30% of the

salt on the total salt used. Mechanical desalting can also be carried out on specially designed brushing machines. After mechanical or normal desalting, the hides and skins are sorted grade-wise and taken for subsequent processes.

3.1.3. Soaking

In this step, the raw skins or hides are washed first to remove the dirt, dung and blood. Then the skins or hides are soaked in water. Additionally, globular proteins as well as salts used for preservation are removed. The process duration is about 8 to 20 hours for wet salted skin and 24 to 48 hours for dry salted and dried skin. The wastewater, which contains salts, chemical additives, and excrements is discharged into a water treatment plant. Soaking is usually carried out in processing vessels (e.g., mixers, drums, or paddles) in two steps, namely a dirt soak for salt and dirt removal, and a main soak for rehydration of hides or skins. The soak bath is often changed every 8 hours to prevent bacterial growth.

3.1.4. Liming

The second step in the leather process is the removal of hair, adipose layer and lipids. The treatment affects the structure of the skin, which results in a better reactivity of the skin containing collagen when it is exposed to

tanning agents. During liming, chemicals such as lime ($\text{Ca}(\text{OH})_2$) and Sodium Sulfide (Na_2S) are used. Liming process opens up the fibre structure of the skin and hence provides more working surface for treatment with tanning agents. Furthermore, natural fats are partially saponified, most of the interfibrillar proteins such as albumins and globulins are eliminated and the mucoids are degraded.

In the pit or paddle unhairing process, 300-400% water, 5-10% lime, and 2 – 4% sodium sulphide are employed, whereas, in the case of paint liming, only 10-15% water is used to make a paste of suitable consistency. After the removal of hair, the goods are re-limed in order to open up the fibre structure suitably, depending upon the type of finished leathers to be made from them or the type of tanning system to be employed.

3.1.5. Re-liming

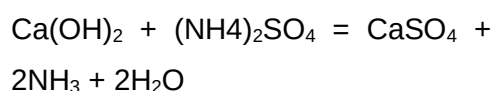
Re-liming either is carried out in pits or paddles mostly with 300- 400% water and lime. Soda ash or sodium hydroxide may be used in small quantities to effect plumping to obtain better opening up of fibre structure in shorter duration. After re-liming, the goods are fleshed and scudded to remove short hair and dirt.

3.1.6. Fleshing

Fleshing is the removal of the adipose layer from the pelt. This operation is performed either manually or by using fleshing machine. This operation results in the generation of the process solid waste 'fleshings'. About 10 kg of fleshing waste is generated from every ton of raw hide/skin.

3.1.7. De-liming

The purpose of de-liming is to remove the lime and other alkalis (free alkali and combined alkali) present in pelt by washing and by using chemicals. Free alkalis can be removed by repeated washings but for removal of combined alkalis demands use common organic acids like lactic acid, acetic acid, formic acid etc. or ammonium salts like ammonium chloride, or ammonium sulphate. The most common de-liming agent used is ammonium sulphate, which reacts with lime as indicated below, forming calcium sulphate. Calcium sulphate formed is subsequently removed during washing.



3.1.8. Bating

Bating is carried out after completion of de-liming with the help of bating enzymes to remove the inter-fibrillary substances leading to cleaner pelt and to engender softness to final

leather. The enzyme used for bating, denatures the inter-fibrillary substance and the same is removed. The effectiveness of bating is checked by air bubble test on skins and by the retention of thumb impression on hides.

3.1.9. Pickling

Pickling is done for acidification of the pelt to bring it to a suitable pH level to carry out tanning. The pH of the pelt after de-liming is around 8 and it needs to be brought down to 2.8 or 3.8 for carrying out chrome tanning or vegetable tanning respectively. For bringing down the pH, mineral acids such as sulphuric acid or hydrochloric acid are used. To suppress swelling due to the use of these acids, common salt is used. Concentration of the float is of immense importance before addition of sulphuric acid as the low salt concentration may lead to acid swelling resulting in poor tear strength and grain bursting strength. High concentration on the other hand, leads to poor substance in final leather. Salt concentration should be checked using Beaume meter and it has to be in the range of 6-7 before addition of acid.

3.1.10. Degreasing

Though a large part of the fat and natural grease is removed during the liming process by saponification, with skins like wool sheepskins, which contain lot of fats, it might become

necessary to carry out degreasing process separately. The process may be carried out either after de-liming or after pickling. The degreasing is usually carried out without float and washing with water in the case of de-limed pelt and with saline water with pickled pelt is done to remove solubilized fat.

3.1.11. Tanning

It is the process of converting putrescible hides and skins to the state of non-putrescible and the mechanical and hydrothermal stability of the skin network is enhanced. The place where tanning process is carried out is known as tan-yard. There are different methods of tanning. However, chrome tanning and vegetable tanning are the popularly practiced tanning methods.

a. Chromium Tanning: The pickled pelts are treated using Basic Chromium Sulphate (BCS). The chromium tanning salt penetrates the skin network and 'coordinate covalent bonds' are formed between the chromium and the carboxyl groups of collagen. The cross links formed brings about the stability of the skin network. The fixation of the chromium is achieved by treating the leather using sodium formate and sodium bicarbonate. In the end,

pH of the leather is increased to 4.0 from the initial pH of 2.8.

- b. Vegetable Tanning:** Vegetable tanning is the other widely adopted method of tanning for commercial production of leather. The astringent organic compounds obtained from the plant kingdom, known as tannins which are capable of converting raw hides/ skins into leather, are used for tanning. The tanning extracts are obtained from wattle, quebracho, chestnut, gambier, myrobalan etc. and used for tanning in either pits or drums. The most important control during vegetable tanning is the specific gravity of the tan liquor, which is measured by using hydrometer called Barkometer. Vegetable tanning usually starts at a specific gravity of the liquor at 1.005 (50 Bk) and the strength of the tan liquor gradually increased as the tanning proceeds.

3.1.12. Post Tanning Processes

The processes that are performed after tanning but prior to finishing such as neutralization, dyeing, re-tanning and fatliquoring are called post tanning or wet finishing operations. The main objectives of this set of unit operations are to render colour to the leather as

demanded by the market (dyeing), impart softness (fatliquoring), fullness, and uniformity of substance (retanning).

The various unit operations carried out during post tanning are provided below:

- a. Wetting Back:** In case, due to improper upkeep some portions get dried up, proper 'wetting back' is needed. For this purpose, water and some surface- active agents or detergents are needed to reduce the surface tension of water so that it can penetrate better and faster to wet back the chrome-tanned leathers.
- b. Re-chroming:** Re-chroming is a process of treating the chrome-tanned leathers with basic chromium sulphate or chrome syntan. Re-chroming is normally done to achieve more uniform chromium content in a batch of leather and uniform layer wise distribution of chrome in thicker leathers. Re-chroming also improves the amenability to dyeing of the leathers.
- c. Neutralization:** Shaved leathers are washed in running water to remove free acid present in leather. Neutralization is carried out to remove the combined acid in leather by using mild alkalies

and other auxiliaries to bring the leather to the right pH condition for subsequent processes by removing excess positive charge. Free acid and acid salts thus liberated after neutralization are removed by washing the leathers alternately in hot and cold water.

- d. **Re-tanning:** It is the process of treating the tanned leathers with synthetic tanning agents. This process is carried out in order to incorporate the required bulk properties in the final leathers. Re-tanning improves the fullness and uniformity of the substance, it also fills the looser areas in order to get uniform substance and fibre compaction in all parts of the skin, imparts body and round feel to the final leather, avoids grain looseness, improves 'break' and grain tightness properties. Re-tanning also improves overall cutting value, buffability, burnishability and print retention properties.
- e. **Dyeing:** Leathers are dyed in order to impart colour as per fashion demand. For leathers like suede and nubuck, dyeing is the final process. Leathers can be surface dyed, through and through dyed or tone in tone dyed as per demand with right selection of dyes. The dyes

should be carefully and completely dissolved before addition into the drum. The dyes before use must also be scrutinized for the presence of banned arylamines.

- f. **Fat-liquoring:** Fatliquoring is done to impart softness, flexibility, feel, drape, run etc. to the leather even after drying. The individual fibres are coated with a thin layer of oil to reduce internal friction and make the fibres slide over one another to give required softness and flexibility. The strength properties are also improved by fatliquoring. Different types of oils for fatliquor preparation are used based on many parameters like molecular size, viscosity, drying characteristics, degree of unsaturation, free fatty acid content etc.
- g. **Fixing:** Synthetic tanning agents, fatliquor and dyes are fixed to the leather by using formic acid.
- h. **Finishing:** Finishing is the last set of operations done in leather processing to improve the appearance, aesthetic and sale values of the leathers and protecting the leather surface from damage during usage against rubbing, water, soiling and abrasion and scuff and

vagaries of weather. The final colour as demanded by the buyer/market is adjusted in finishing. An attempt is also made to hide the surface blemishes aiming at the upgradation of the leathers. In finishing, special surface effects and modification of texture are also achieved. The main objectives of finishing are summarized as follows:

- Protecting it from damage by water, soil and mechanical action
- Improving its physical properties such as its lightfastness and rub fastness
- Levelling out patches and grain faults (upgradation)
- Applying an artificial grain layer to splits or corrected grain leathers
- Modifying the surface properties (shade, lustre, handle, etc.)

'Finishing season' for use in a process called transfer coating is prepared by using pigments, dye solutions, binders, wax emulsion, fillers and other auxiliaries is applied on leather surface by either padding, spraying or roller coating. Base coats are given for uniform distribution and adhesion of

finishing season. Nitro cellulose lacquers or polyurethane based top coats are provided to protect and improve performance properties of the finish film. Foam finishing is a new concept where better covering is obtained especially on upholstery leathers.

3.2. Mechanical Operations

3.2.1. Sammying

Sammying is the process of removing excess water from wet blue hides and skins to prevent damage during splitting/ shaving. In the tanneries it is done by sammying machine or setting machine. The wet blue hides and skins are well spread and squeezed between felts to reduce the moisture content to nearly 50-55%. This operation also helps in removal of folds and wrinkles from the wet blue hides and skins.

3.2.2. Splitting

Splitting is an operation carried out to split the material (leather) into two or more layers to the desired thickness. Splitting is normally done after tanning. In the case of hides (i.e., cow and buffalo), the thickness of the material while processing is more compared to the thickness required for the final leather. The extent of splitting to be carried out depends upon the thickness required in the final leather. The set splitting thickness will be normally a little more

than that is desired in the final leather. This allowance is given because, due to the variation in the thickness of the material, it is not possible to get a material of uniform thickness after splitting, as there is shrinkage in the thickness while processing. Generally, an allowance of 20-40% on the final thickness is given.

3.2.3. Shaving

The object of shaving is to make the thickness uniform all over the hide and the shaving is done only on the thicker part of the leather.

3.2.4. Setting

After post tanning process, the leathers are washed and piled in the yard. Next day, the crust leathers are taken for setting operation. Setting is carried out to make the leather flat and smooth and removes the wrinkles before going for vacuum dry or overhead chain drive. Setting operation improves the area yield.

3.2.5. Drying

Drying is the process to remove the water from leather. After the setting operation, the crust leathers still contain some amount of water. In order to carry out the subsequent mechanical operations, crust leather has to be in completely dried condition. The Drying operation is generally carried out in tannery to bring the leather to a practically usable form. Therefore, drying is

physical as well as chemical activity and is of great importance to the practical quality of leather. Types of drying process and rate of drying influence the final properties of crust leathers, for example, softness. If the drying process is carried out rapidly, the final leather become stiff, cracky, and horny and may have boat shape, which results in area shrinkage. So, the subsequent mechanical operations, for example, staking and toggling operation may be difficult. In addition, very slow dried leathers generally show fungus and molds on leather surfaces, especially vegetable tanned leathers.

- a. **Suspension Drying:** It is a method of drying where leather is simply hung up in the drying room or tunnel, which has ions of heating, humidity and air circulation. It is also called hang drying which indeed is an example of conventional drying method.
- b. **Vacuum Drying:** The most significant advancement in the drying technique applied to the leather is the development of the vacuum dryer. It is this principle that makes vacuum dryer come into use, which reduces the time of drying and to retain those properties of the drying materials, which are susceptible to temperature.

Vacuum drying is becoming increasingly popular of late for most full grain, aniline leathers whilst it is also perfectly satisfactory for corrected grain types. In vacuum drying, the set-out leathers are laid flat and grain side down on to smooth stainless-steel table where they are also stretched out firmly by hand slickers as their shape dictates. Heat is applied to this surface by a built-in head exchanger under the table. This temperature is maintained by thermostatic control of circulating hot water, and a hood is placed over the plate and then evacuated to aid in drying the leather. Since the heat is supplied directly to the skins, evaporation will take place in the fibres inside the leather, and the problems arising from migration of the tanning materials and oils are greatly decreased.

- c. **Toggle Drying:** A toggling unit consists of a number of screens placed in a dryer having controlled temperature and humidity. The leather is stretched and held in place by a number of toggles that look into the screens. Advantages of toggle drying are:

- Drying the large quantities of leather in a relatively small space
- Provides stretching to the leather during drying. There are some disadvantages like constant temperature and humidity conditions are difficult to maintain and control.

3.2.6. Conditioning and Staking

Conditioning is carried out by spraying water on the surface of crust leathers. The water sprayed leathers are aged for an hour and then taken for staking operation. This is carried out to make the leather uniformly damp for staking. The purpose of staking is to soften the leather. Pressure is adjusted to produce a soft and smooth grain and feel of final leather. If the pressure is too high, then we get loose and pipy grain.

3.2.7. Buffing

The leather is buffed on the flesh side to remove the loose bits of flesh and give a velvety feel. The colour on the flesh side also becomes uniform. The buffing is done on the flesh side for all types of leather to remove the loose fibres by emery papers. The purpose of buffing is simply to improve the hide appearance by dressing or refurbishing the grain or the flesh side.

3.2.8. Spraying Machine

Spraying machines are used instead of hand spraying for bulk production. The flat leather is fed on to a nylon band conveyor, travelling at speeds of 40- 70 ft. per minute, and passes beneath the spray guns. One or more spray guns may be used. In a rotary spraying machine (Auto Spray), the guns are mounted on the arms of a rotor, which rotates above the conveyor band. Once again, the spraying of the guns is limited to the period of time when leather is actually beneath them, by mounting sensing elements on the rotor arms so that they will sense the incidence of leather beneath them just before it comes beneath the adjacent spray gun. These sensing elements may be of the photoelectric eye type or the ultrasonic beam type.

3.2.9. Plating/Embossing

These are carried out using a hydraulic press. Machine plating is done at relatively lower temperature but with more pressure than ironing, which is done at a higher temperature, but with momentary pressure. Machine plated leathers become somewhat firmer and stiffer because of compression. Vegetable and re-tanned leathers are more susceptible in this respect. For producing a milder handle, higher temperature with less pressure is advisable. In smooth plating, the

plate is a highly polished one; the more the polish, the smoother the leather surface is obtained. Printing and embossing differ from smooth plating; engraved or etched plates of several designs are used for embossing. Vegetable and re-tanned leathers take a deeper print when compared to chrome leathers. A longer lasting grain is obtained when the leather is dampened a little before embossing. On chrome leather, a fairly durable print is possible only by using high temperature, pressure and dwelling time.

3.2.10. Measuring

Once the leathers are finished, they are measured using a measuring machine. The measurement of leathers is carried out at wet blue stage and crust stage.

3.2.11. Packaging

One leather is kept on a flat surface with grain side up, then 10 to 12 pieces of leathers are placed over it keeping flesh side up and rolled into bundles. Further, it is wrapped in polyethylene sheets to prevent regaining moisture, protect leathers from molds, fungus, and mildew growth.

- Finished leather (glazed and patent leather) can be packed in poly ethylene plastic bags and/or wrapped with jutes, sisals etc., which is used to protect the gloss from fading.

- Resin finished leather can be packed with normal paper.

3.3. Leather Products

3.3.1. Footwear (Shoes, Slippers, Sandals)

A shoe is an item of footwear intended to protect and comfort the human foot while carrying out various activities. Shoes are also used as an item of decoration. The design of shoes has varied enormously through time and from culture to culture, with appearance originally being tied to function. Additionally, fashion has often dictated many design elements, such as whether shoes have very high heels or flat ones. Contemporary footwear varies widely in style, complexity and cost.

General classification of Footwear:

- Formal shoes
- Dress Casual Shoes
- Boots
- Sandals
- Pantafola
- Sports footwear
- Therapeutic Footwear
- Speciality footwear (Safety shoes etc.)

Generally, footwear making is carried out in the following stages:

- PED (Product engineering & development – where designing & Pattern making and samples been made)
- Clicking
- Pre-closing

- Closing
- Assembly
- Finishing

a. Making the Shoe Last: The first step in shoemaking is making of a shoe last is a mould that emulates a foot to give the shoe its shape. Traditionally, these were carved from wood but now plastic and metal are also used. Every left and right shoe needs a last so its shape and size can be determined. A number of things are taken into consideration while preparing a last, which include how a foot rolls when one walks and how this will affect factors such as heel height. Later in the shoemaking process, the last is placed inside the footwear so it can be modelled around it. This is used again once the shoe is almost finished to make sure the end fit matches the original design/basic footwear pattern. To make a shoe pattern one would need to attach masking tape on the shoe last and apply the shoe design on the shoe last. Then the shoe masking tape pattern is taken off the shoe last and attached to a cardboard and then finally cut out. Made from the basic shoe pattern, the upper and lining

pattern to each line of the basic shoe pattern, adds allowance.

- b. Clicking:** The production process starts with cutting the materials. Steel dies in a hydraulic press generally cut the components. The cutting dies for Footwear components look like simple cookie cutters but they are very sharp. Every component requires its own cutting die. Every part for every size needs its own die. These shoe parts are called the shoe pattern. To support the production of high-volume shoes in popular sizes, often many die sets are required. The cutting dies are placed on the leather or fabric materials by hand, and then the press closes to make the cuts.

After the parts are cut, a worker will carefully organize and stack the components. The parts will then be moved on to be processed before stitching. The parts will be marked for alignment or have logos added by the embroidery or printing departments. Once the parts are prepared, the workers will organize the parts into kits for the stitching department.

- c. Making of the upper (Closing Operation):** Once the components are processed and

organized, the stitching process is carried out which may be divided into simple steps, especially in bigger set-ups. Each worker is given one task to complete. For example, one worker will sew the vamp parts to the mudguard then pass the shoe onto the next worker to attach the eye stay parts. With each operation assigned to one worker, the QC staff can quickly track any problems. The division of the operations also allows the worker to quickly master their operation. In the process, the upper parts are assembled, the lining is attached and the foam for collars and tongues are inserted.

The upper is completely assembled with the reinforcing parts added and heat moulded into shape. Toecaps and heel counters are formed and ready for the final assembly operations.

- d. Assembly Process:** With the stitching complete, the upper is ready to be bonded to the outsole. Before the outsole can be attached, the upper must be “lasted.” The assembly and finishing process comprises broadly of the following steps:

- **Lasting:** Lasting is a process during which you put the shoe upper on the shoe last and attach the upper to the insole. The purpose of lasting shoes is to stretch the shoe upper on the shoe last and to give it the desired shape. Lasting shoes involve different shoemaking techniques working with shoemaking pincers, shoemaking nails and contact glue.
- **Surface preparation:** With the upper tight to the last, the outsole unit can be permanently cemented to the upper. During this process, the shoe travels inside a heat tunnel to dry the glue before the final bonding process.
- **Sole Attaching:** After the step of shoe lasting, sole is attached. There are few materials in which the soles are made and the most common material is vegetable-tanned leather for leather soles, TPR sole, TPU soles, Rubber soles, EVA soles etc. Appropriate sole should be attached based on the design requirement and last profile.
- **Shoe finishing:** After de-lasting, the socks or foot insert is attached to the footwear and then aesthetic appeal is given to the footwear by brushing, applying creams to the upper or spraying technique etc. There are many types of finishes available; some popular finishes are mirror finish, brush off finish, oil and burnishable finish. Finally, the footwear is checked for quality and then packed.



Figure 2: Preparation of the “Last”



Figure 3: Finished product

3.3.2. Jutti/Jooti

Apart from being called as 'Juttis' or 'Mojari' they are also called 'Khussa' in Punjab and are very popular by that name in western part of India. These shoes are used both during festivities and as day-to-day use. The juttis worn on joyful events are rather elaborate and decorated with gold threads and other embellishments while the day-to-day wear Juttis are more utilitarian, and lack the fancy embellishments. They are not made for the conventional right and left foot. The Jutti's take the shape of the foot in which they are worn for a period.



Figure 4: Jutti Upper

The parts of Juttis are:

- **Leather sole:** The sole is cut from a leather strip, is shaped, and smoothened using a steel brush or a similar abrasive.
- **Shoe Upper:** This is the upper part of the Jooti and is often covered with embellishments. The artisans embroider details or make patterns with perforating

the uppers in various floral patterns.

a. Manufacturing

- The edges of the upper are folded inwards and the sole is glued to the upper.
- Using an awl and cotton thread the artisan's stitch the shoe upper and the sole along the inward folded edges. This does not leave any visible threads on the outer side of the shoe.
- Any loose threads are carefully removed.

3.3.3. Saddlery

The saddle is a supportive structure for a rider of an animal, fastened to an animal's back by a girth. The most common type is the equestrian saddle designed for a horse. A modern saddle has the following parts:

- **Tree:** It is the base on which the saddle is built. It is usually made of wood or a similar synthetic material. The tree's size determines its fit on the horse's back, as well as the size of the seat for the rider and it provides a bearing surface to protect the horse from the weight of the rider. The solid saddle tree raises the rider above the horse's back and distributes the rider's weight on the horse's back, thus greatly increasing the comfort of the

horse and prolonging its useful life.

- **Seat:** the part of the saddle where the rider sits, it is usually lower than the pommel.
 - **Pommel or pommel (English)/ swells (Western):** the front, slightly raised area of the saddle.
 - **Cantle:** the rear of the saddle
 - **Stirrup:** part of the saddle in which the rider's feet are placed; provides support and leverage to the rider.
 - **Leathers and flaps (English), or fenders (Western):** The leather straps connecting the stirrups to the saddle tree and leather flaps giving support to the rider's leg and protecting the rider from sweat.
 - **D-ring:** a "D"-shaped ring on the front of a saddle, to which certain pieces of equipment (such as breastplates) can be attached.
 - **Girth or cinch:** A wide strap that goes under the horse's barrel, just behind the front legs of the horse that holds the saddle on.
 - **Panels, lining, or padding:** Cushioning on the underside of the saddle
- a. **Manufacturing**
- **Making of the Tree:** The different pieces, which comprise the Tree, are cut and fashioned individually. These are assembled and

coats of Polyurethane are applied to protect the wood from the raw hide.

- **Raw Hide covering:** The saddle is covered in rawhide (animal skin that has not been tanned). This stiffens as it dries wrapped around the tree frame.
- **Cutting the leather:** The Leather is cut in shapes and wetted. The Tree and cantle are covered in leather held in place with nails.
- **Fitting the seat:** The seat is fixed on the saddle and shaped. Sanding is done to achieve a smooth finishing.

3.3.4. Leather belt

A leather belt is a fashionable accessory and sometimes necessary addition to a pair of pants. Leather belts are usually made from vegetable tanned hides, as they are stiffer than chromium tanned dyes. The process of making a belt can be described below as:

- **Size:** The belt size is determined and accordingly the leather strip is cut
- **Marking the placement holes:** The holes are marked on the leather strip and punched
- **Attaching the Belt loop and the buckle**
- **Trimming the edge of the belt**
- **Application of dyes on the belt**

3.3.5. Leather bag

A handbag, commonly known as a purse in North American English, is a handled medium-to-large bag used to carry personal items. It has become a part of men as well as women's daily use articles. The process of making a handbag can be described below as:

- **Size:** The leather is cut into strips and other shapes for the bag. Edges that will receive multiple layers of the leather.
- **Stitching the Sides:** The leather is folded into half. A gusset is created (a W shape fold at the bottom of the bag) and the sides are stitched.
- **Turning out:** The stitched bag is turned inside out and the W shape smoothened out. Handles and zippers are attached to the bag.

3.3.6. Leather wallet

A leather wallet is an accessory used for carrying money. It is one of the most easily recognized accessory and is also used to carry several documents which are required to be produced at a short notice such as ID cards, Licenses etc. The process of making a leather wallet can be described below:

- **Sizing:** The size of the leather is cut as per the size of the wallet to be made.
- **Marking:** The marking is made for the fold lines on the fuzzy side

of the leather and a bone folder is run along the line.

- **Clamping:** The folds are folded in and held in place using clamps to set the shape.
- **Punching holes:** Using a special tool, sewing holes are marked on the inside pockets. Now the wallet is turned over and using an awl the holes are punched in over the slightly raised indentations.
- **Sewing:** The wallet pockets are sewn together using a strong thread and the extra threads are removed.

3.3.7. Leather jewellery

Leather jewellery primarily takes the form of bracelets or earrings and other traditional jewellery items like pendants, key chains etc. made out of leather. The simplest way could involve braiding the leather strips.

The process may be described as below:

- **Cutting the leather:** A soft leather (chrome tanned) is used and strips are cut from it.
- **Braiding:** Braiding may be done using the three-strand braid. The top is secured and the left strand is placed over the central strand. The next step involves moving the right strand over the central strand. This is also referred to as the hair braiding technique.

Now the braided leather can be attached together to form a bracelet,

or attached to a key chain ring to form a key chain.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for uniformity of the stitches
- Check the leather for any discoloured patches
- The Holes punched should be equally spaced (for belts)

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided

below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	Quality Control Order, (2021) For Leather and Footwear Sector	BIS	Specifies mandatory certification and Quality mark for leather and leather footwear

5.2. International

S. No.	Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) regulation (EC) 1907/2006	a. European Commission b. European Chemicals Agency (ECHA)	a. Specifies the duty of the importers and manufacturers in ensuring that the product is not posing threat to customers b. Specifies regulations on Heavy metals and hazardous chemicals
2	United States of America	a. California Proposition 65, Act of 1986	a. California State	a. Any manufacturing unit having more than ten employees are required to

S. No.	Country	Regulation	Regulatory Body	Remarks
3		b. Consumer Product Safety Improvement Act (CPSIA), Act of 2008	b. Consumer Product Safety Commission (CPSC)	label products if they contain carcinogenic products b. Specifies requirements with respect to Chemicals and heavy metals and tracking label c. Specifies labelling and advertisement claims
	Australia	a. Labelling Requirements	a. Australian Competition and Consumer Commission (ACCC)	a. Specifies labelling requirements for clothing including leathers
4	Middle East	a. Saudi Standards, Metrology, and Quality Organization (SASO) regulations	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Specifies technical regulations for SASO and type of certification (1A) which mandates product testing

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 1640: 2007 (Reaffirmed Year: 2017)	Glossary of terms relating to hides, skins and leather	Australia	ISO 15115: 2019	Leather: Vocabulary
Raw Materials (Refer Table 2.a & 2.b)	IS 579: 2017	Vegetable tanned sole leather - Specification (Fourth Revision)	Middle East	GSO 2442: 2014	Leather - Tanned leather with chrome (wet blue) used in different leather industry
	IS 578: 1985 (Reaffirmed Year: 2019)	Full-Chrome upper leather	Middle East	GSO EN 12203: 2011	Footwear, leather and imitation leather goods manufacturing machines - Shoe and leather presses
	IS 582: Part 5: Section 1: 2018	Part 5 : Sec 1 : 2018, methods of chemical testing of leather, part 5 determination of certain azo colorants in dyed leathers Section 1 Determination of certain aromatic amines derived from azo colorants	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 2954: 1978 (Reaffirmed Year: 2017)	Vegetable tanned leather for belting	-	-	-
	IS 11944: 1987 (Reaffirmed Year: 2019)	Specification for wallet leather	-	-	-
	IS 6191: 1971 (Reaffirmed Year: 2019)	Micro-Biological colour fastness and microscopical tests for Leather	-	-	-
	IS 14583 : 1998 (Reaffirmed Year: 2015)	Specifications for suede leather lining	-	-	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 1989: Part 1: 1986 (Reaffirmed Year: 2021)	Leather safety boots and shoes - Part 1 : For miners	European Union	BS EN ISO 20137: 2017	Leather-Chemical tests : Guidelines for testing critical chemicals in leather
	IS 5867:1970 (Reaffirmed Year: 2019)	Leather boards for insoles	Australia	ISO 17131: 2020	Leather - Identification of leather with microscopy
	IS 5677: 1986 (Reaffirmed Year: 2019)	Shoe upper leather for direct moulding processes	Australia	ISO 17075-2: 2017	Leather - Chemical determination of chromium(VI) content in leather
	IS 17037: 2018	Anti-riot shoes, specifications	Middle East	GSO ISO 17227: 2013	Leather - Physical and mechanical tests - Determination of dry heat resistance of leather
	IS 17012: 2018	High ankle tactical boots with PU — rubber sole — specification	Middle East	GSO ISO 4044: 2014	Leather: Chemical Tests - Preparation Of Chemical Test Samples
Finishing (Refer Table 4.a & 4.b)	IS 582: 1970	Chemical testing of leather, methods of sampling	European Union	BS EN ISO 20433: 2012	Leather: Tests for colourfastness. Colour fastness to crocking
	IS 5914: 1970 (Reaffirmed Year: 2019)	Methods of physical testing of leather	European Union	BS EN ISO 4044: 2017	Leather: Chemical tests. Preparation of chemical test samples
	IS 17011: 2018	Chemical requirements for footwear and footwear materials	European Union	BS EN ISO 11642: 2012	Leather: Tests for colourfastness. Colour fastness to water
	-	-	USA	ASTM D1610-18	Standard practice for conditioning leather and leather products for testing

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
	-	-	USA	ASTM D5552 - 10 (2015)	Standard test method for resistance of coloured leather to bleeding
	-	-	USA	ASTM D4705-18	Standard test method for stitch tear strength of leather, double hole
	-	-	USA	ASTM D2098-13	Standard test method for dynamic water resistance of shoe upper leather by the Dow Corning leather tester
	-	-	Australia	ISO 11410: 2021	Leather -Guidelines for packaging of wet blue leather

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 1640: 2007(Reaffirmed Year: 2017) - Glossary of terms relating to hides, skins and leather	This standard defines the terms strictly relating to hides, skins and in the Indian leather trade and industry
	ISO 15115: 2019 - Leather: Vocabulary	This document defines terms used in the leather industry

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 579: 2017 - Vegetable tanned sole leather - Specification (Fourth Revision)	This standard prescribes requirements and the methods of sampling and tests for vegetable tanned sole leather including those which have been specially treated for water-resistance property
	IS 578: 1985 (Reaffirmed Year: 2019) - Full chrome upper leather	- This standard prescribes the requirements, methods of sampling and test for full-chrome upper leather. - It does not cover the requirements of upper leather manufactured by tannages other than full-chrome, and excludes mouldable and patent leather uppers
	IS 582: Part 5: Section 1: 2018 - Methods of chemical testing of leather, part 5: Determination of certain azo colorants in dyed leathers, Section 1 Determination of certain aromatic amines derived from azo colorants	- This standard prescribes the methods of chemical testing for all types of leathers - Should there be any inconsistency between the methods prescribed in this standard and those given in the individual material specification, then the latter shall prevail
	IS 2954: 1978 (Reaffirmed Year: 2017) - Vegetable tanned leather for belting	This standard prescribes the requirements and the methods of sampling and test for vegetable tanned leather used for flat and round leather beltings

Stage	Code	Scope
	IS 11944: 1987 (Reaffirmed Year: 2019) - Specification for wallet leather	This standard covers requirements for leather wallet
	IS 14583 : 1998 (Reaffirmed Year: 2015) - Specifications for suede leather lining	- Chemical requirements for suede leather lining - Physical requirements for Suede leather lining

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	GSO 2442-2014 - Leather - tanned leather with chrome (wet blue) used in different leather industry	This GSO standard specifies general requirements. Physical and chemical properties for wet blue leather either with wool or hair (fur) or without wool or hair (fur) used in varies leather industries such as clothing , belts, gloves etc.
	GSO EN 12203: 2011 - Footwear, leather and imitation leather goods manufacturing machines - Shoe and leather presses	This European Standard is applicable to shoe and leather presses (see 3.1) used in the manufacture of footwear, leather and imitation leather goods and other related components. These machines are: - Sole attaching presses (open and closed types); - Sole and insole moulding machines; - Back part moulding machines; - Backer, lining and toe puff attaching presses

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 1989: PART 1: 1986 - Leather safety boots and shoes - Part 1: For miners	This standard (Part 1) prescribes the requirements and methods of sampling and test for leather safety boots and shoes foremen miners
	IS 5867:1970 (Reaffirmed Year: 2019) - Leather boards for insoles	This standard prescribes the requirements and the methods of sampling and test for leather boards for footwear insoles
	IS 5677: 1986 (Reaffirmed Year: 2019) - Shoe upper leather for direct moulding processes	This standard prescribes the requirements, and methods of sampling and test for full-chrome tanned upper leathers used for direct moulding processes
	IS 17037: 2018 - Antiriot shoes, specifications	- Physical requirements (Size, Weight etc.) - Referenced documents and adjunct standards - Definition of defects - Tests
	IS 17012: 2018 - High ankle tactical boots with PU — rubber sole — Specification	- Physical requirements (Size, Weight etc.) - Referenced documents and adjunct standards - Definition of defects - Tests

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	BS EN ISO 20137 – 2017 - Leather: Chemical tests. Guidelines for testing critical chemicals in leather	- This document gives guidelines to apply the available chemical test methods for leather. This information can be used by those involved in setting specifications for leather, especially for those parameters relating to restricted chemical substances - Lists of restricted chemicals contain many substances that are not relevant to the leather industry. Those chemical substances that are not mentioned in this document do not need to be determined, thus avoiding unnecessary analytical costs
	ISO 17131: 2020 - Leather: Identification of leather with microscopy	Leather - Identification of leather with microscopy
	ISO 17075-2: 2017 - Leather physical and mechanical tests - Determination of dry heat resistance of leather	This document specifies a method for determining chromium (VI) in solutions leached from leather under defined conditions. The method described is suitable to quantify the chromium (VI) content in leathers down to 3 mg/kg.
	GSO ISO 17227: 2013 - Leather - Physical and mechanical tests - Determination of dry heat resistance of leather	This International Standard specifies a method of determining the dry heat resistance of conditioned leathers. It is applicable to all leathers
	GSO ISO 4044: 2014 - Leather - Chemical tests - Preparation of chemical test samples	This document specifies how to prepare a test sample of leather for chemical analysis. The test sample can be either ground or cut into small pieces. Unless specified in this document, the method to be used depends on the size of leather sample available for testing

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 582: 1970 - Chemical testing of leather, methods of sampling	- This standard prescribes the methods of chemical testing for all types of leathers - Should there be any inconsistency between the methods prescribed in this standard and those given in the individual material specification, then the latter shall prevail
	IS 5914: 1970 (Reaffirmed Year: 2019) - Methods of physical testing of leather	Methods of physical testing of leather
	IS 17011: 2018 - Chemical requirements for footwear and footwear materials	- Requirements and methods of sampling - Methods of Testing - Chemicals for information purpose (for which no tests were available at time of publication of standard) - Referenced documents and adjunct standards

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	BS EN ISO 20433 – 2012 - Leather: Tests for colourfastness. Colour fastness to crocking	- This International Standard specifies a method for determining the amount of colour transferred from the surface of coloured leather to other surfaces by rubbing with a white cotton cloth - Two tests are carried out, one with a dry rubbing cloth and one with a wet rubbing cloth
	BS EN ISO 4044 – 2017 - Leather: Chemical tests. Preparation of chemical test samples	This International Standard specifies a method for the preparation of a test sample of leather for chemical analysis. This method is applicable to all types of leather
	BS EN ISO 11642: 2012 - Leather: Tests for colourfastness. Colour fastness to water	This International Standard specifies a method for determining the colourfastness to water of leather of all kinds at all stages of processing
	ASTM D1610-18 - Standard practice for conditioning leather and leather products for testing	- This practice covers the conditioning of all units and specimens of leather and leather products prior to testing and the conditions under which they should be tested. This practice does not apply to wet blue - This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use
	ASTM D5552-10(2015) - Standard test method for resistance of coloured leather to bleeding	- This test method covers the determination of the resistance of coloured leather to bleeding. This test method does not apply to wet blue. - This test method is intended for use on finished upper leather for shoes or similar end use, or the leather article such as shoe, handbag, etc.
	ASTM D4705-18 - Standard test method for stitch tear strength of leather, double hole	- This test method is intended for use in determining the stitch tearing resistance of leather using a double hole tear. It is particularly applicable to lightweight leathers. This test method does not apply to wet blue. - This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use
	ASTM D2098-13 - Standard test method for dynamic water resistance of shoe upper leather by the Dow-Corning leather tester	This test method covers the determination of the dynamic water resistance of shoe upper leather by means of the Dow Corning Tester. It has been revised to show the state of the art in the equipment used in testing. It is applicable to all types of shoe upper leather. This test method does not apply to wet blue
	ISO 11410: 2021 - Leather - Guidelines for packaging of wet blue leather	Leather - Guidelines for packaging of wet blue leather

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information: <http://odopup.in/en/article/kanpur-nagar>
2. Product information: <http://odopup.in/en/article/Agra>
3. Product information: <http://odopup.in/en/article/Unnao>
4. Product information: <http://odopup.in/en/article/Hamirpur>
5. Product information : https://en.wikipedia.org/wiki/Leather#From_other_animals
6. Product information : https://en.wikipedia.org/wiki/Leather_production_processes
7. Product Information: [https://en.wikipedia.org/wiki/Tanning_\(leather\)](https://en.wikipedia.org/wiki/Tanning_(leather))
8. Product Information: <https://italianshoefactory.com/blog/how-shoes-are-made/>
9. Product Information: <https://en.wikipedia.org/wiki/Saddle>
10. Product Information: [https://en.wikipedia.org/wiki/Rawhide_\(material\)](https://en.wikipedia.org/wiki/Rawhide_(material))
11. Product Information: https://www.freckerssaddlery.com/saddle_tree.php
12. Product Information: <https://www.instructables.com/How-To-Make-a-Leather-Bag/>
13. Product Information: <https://www.craftpaperscissors.com/how-to-make-leather-jewelry/>
14. Mandatory IS mark: <https://bis.gov.in/index.php/product-certification/products-under-compulsory-certification/scheme-i-mark-scheme/>
15. Indian Standards under Compulsory Certification : <https://www.bsbedge.com/mandatory-indian-standards>
16. International Legislation/Regulation:
 - a) EU:
 - <https://www.compliancegate.com/leather-products-regulations-european-union/>
 - <https://www.sherpa-certification.com/en/ce-marking/general-product-safety-gpsd>
 - https://ec.europa.eu/growth/sectors/fashion/textiles-clothing/legislation_en
 - b) USA : <https://www.compliancegate.com/united-states-leather-product-regulations/>
 - c) AUS: <https://www.compliancegate.com/australia-new-zealand-clothing-textiles-regulations/>
 - d) GULF (GCC): <https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation/>
17. Domestic Standards for Leather and leather products: <https://standardsbis.bsbedge.com/>
18. NABL accredited laboratories : <https://nabl-india.org/> , <https://parakh.ncog.gov.in>
19. ISO Standards: <https://www.iso.org/obp/ui#search>
20. International Standards:
 - a) EU : <https://www.en-standard.eu/>, <https://www.bsbedge.com>
 - b) USA: <https://www.en-standard.eu/astm-standards/> , <https://www.astm.org/>
 - c) AUS: <https://www.standards.org.au/>
 - d) GULF: <https://www.gso.org.sa/store/>

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials,

moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products,

services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such

products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL):** <https://nabl-india.org/>
- **Bureau of Indian Standards (BIS):** <https://bis.gov.in/>

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an

organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the

relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National

Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
<http://nabcb.qci.org.in/>

APPENDIX II: IMPORTANT INSTITUTIONS

S. No.	Institute	Areas of Expertise	Postal Address	Contact
1	Central Leather Research Institute (CLRI)	- Leather Process Technologies - Leather Product Technologies - Transfer of Technology	CSIR-Central Leather Research Institute Sardar Patel Road, Adyar Chennai 600 020 India	Email: director@clri.res.in Phone: +91 44 2491 0897 Website: https://www.clri.org/
2	Council for Leather Export (CLE)	- Issue of CLE certificate for Exemption from execution of bank guarantee under Advance Authorization, Duty Free Import Authorization & Export Promotion Capital Goods Schemes - MSME Funding Support	Council for Leather Exports No.1, Sivaganga Road, Nungambakkam, Chennai – 600034	Email: cle@cleindia.com Phone: +044 48684380 Website: https://www.clri.org/
3	Footwear Design & Development Institute, Noida (FDDI)	- Testing Services - Design Services - Consultancy & Project Management Services	Footwear Design & Development Institute, A 10/A, Sector 24, Noida, 2-01301	Email: contact@fddiindia.com Phone: 0120-4500100 Website: https://www.fddiindia.com/
4	Footwear Design & Development Institute, Fursatganj	- Testing Services - Design Services - Consultancy & Project Management Services	Footwear Design & Development Institute, Rae Bareli – Sultanpur Road, Near Police Station, Amethi, 229302	Email: fursatganjcampus@fddiindia.com Phone: 0535-2441534 Website: https://www.fddiindia.com/

